

Cyber Security Minor Project: Kali Linux OS

Name: JAYDEEP JOSHI

Email ID: joshijaydeep348@gmail.com

1. Introduction

Kali Linux is a operating system that is build for cyber security and penetration testing. It have many tools which is very usefull for hackers and security expert. It is essential in cybersecurity domain because it give us safe place to test vulnerability without risk.

2. Objective

The main goals for this minor project is to successfully installing Kali Linux in a virtual environment. Also, we will performing system updates to keep the machine secure from known bugs. After that, we doing MAC address spoofing to hide our device identity on the network, and finally manage user accounts for better security and access control.

3. Setup

For the hardware prerequisites, the host machine need a minimum of 8GB RAM, at least 20GB of free hard disk space, and a multi-core processor with virtualization enabled in the BIOS. For software, we are using Oracle VirtualBox. We need virtualization because it provide a safe, isolated container. If we make mistake and crash the Kali system during testing, our main host operating system will not getting damaged. Virtualization is needed because we not want to damage our main host computer, so tools like VirtualBox help to make isolated virtual environment.

4. Tasks

Installation

I download pre-build VirtualBox image of Kali linux from official site. Then I import it in VirtualBox, set network to NAT, and start it. The default login user is kali/kali.

Start backup

Documents

Kali

kali-linux-2026.1-virtualbox-amd64

Sort

View

Name	Date modified	Type	Size
Logs	13-04-2026 23:15	File folder	
kali-linux-2026.1-virtualbox-amd64	13-04-2026 23:18	VirtualBox Machin...	6 KB
kali-linux-2026.1-virtualbox-amd64.vbox...	13-04-2026 23:13	VBOX-PREV File	6 KB
kali-linux-2026.1-virtualbox-amd64	13-04-2026 23:18	Virtual Disk Image	1,61,00,673...
kali-linux-2026.1-virtualbox-amd64-1.16	20-03-2026 12:42	VirtualBox Machin...	3 KB

Oracle VM VirtualBox Manager

FileMachineHelp

NewOpenSettingsDiscardShow

kali-linux-2026.1-virtualbox-amd64

Running

Details

Preview

General

Name: kali-linux-2026.1-virtualbox-amd64

Operating System: Debian (64-bit)

System

Display

Storage

Controller: IDE

IDE Secondary Device 0: [Optical Drive] Empty

Controller: SATA

SATA Port 0: kali-linux-2026.1-virtualbox-amd64.vdi (Normal, 80.09 GB)

Audio

Host Driver: Windows DirectSound

Controller: ICH AC97

Network

Adapter 1: Intel PRO/1000 MT Desktop (NAT)

USB

USB Controller: OHCI

Device Filters: 0 (0 active)

Shared folders

None

Description

Kali Rolling (2026.1) x64

2026-03-20

Username: kali

Password: kali

* Kali Homepage:

<https://www.kali.org/>

* Kali Documentation:

<https://www.kali.org/docs/>

* Kali Tools:

<https://www.kali.org/tools/>

* Forum/Community Support:

<https://forums.kali.org/>



Update and Upgrade

To update system I open terminal and type command:

```
sudo apt update && sudo apt upgrade -y
```

This download and install latest packages.

```
Session Actions Edit View Help
(kali@kali)-[~]
$ sudo su
[sudo] password for kali:
(root@kali)-[/home/kali]
# sudo apt update
Get:1 http://kali.download/kali kali-rolling InRelease [34.0 kB]
Get:2 http://kali.download/kali kali-rolling/main amd64 Packages [21.1 MB]
Get:3 http://kali.download/kali kali-rolling/main amd64 Contents (deb) [53.5 MB]
Get:4 http://kali.download/kali kali-rolling/contrib amd64 Packages [118 kB]
Get:5 http://kali.download/kali kali-rolling/contrib amd64 Contents (deb) [275 kB]
Get:6 http://kali.download/kali kali-rolling/non-free amd64 Packages [185 kB]
Get:7 http://kali.download/kali kali-rolling/non-free amd64 Contents (deb) [893 kB]
Get:8 http://kali.download/kali kali-rolling/non-free-firmware amd64 Packages [14.3 kB]
Get:9 http://kali.download/kali kali-rolling/non-free-firmware amd64 Contents (deb) [33.8 kB]
Fetched 76.2 MB in 20s (3,858 kB/s)
563 packages can be upgraded. Run 'apt list --upgradable' to see them.
```


3. Change the MAC to a random one: `sudo macchanger -r eth0`

4. Bring interface up: `sudo ifconfig eth0 up`

```
(root@kali)-[/home/kali]
# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:8a:35:d2 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0
        valid_lft 85375sec preferred_lft 85375sec
    inet6 fd17:625c:f037:2:e6c:71f7:75f4:51ca/64 scope global dynamic noprefixroute
        valid_lft 86308sec preferred_lft 14308sec
    inet6 fe80::d650:66de:e6dc:7b4a/64 scope link noprefixroute
        valid_lft forever preferred_lft forever

(root@kali)-[/home/kali]
#
```

```
(root@kali)-[/home/kali]
# sudo ifconfig eth0 down
```

```
(root@kali)-[~]
# sudo ifconfig eth0 down

(root@kali)-[~]
# sudo macchanger -r eth0
Current MAC: 08:00:27:8a:35:d2 (CADMUS COMPUTER SYSTEMS)
Permanent MAC: 08:00:27:8a:35:d2 (CADMUS COMPUTER SYSTEMS)
New MAC: 1a:e2:9a:4d:f8:ec (unknown)

(root@kali)-[~]
# sudo ifconfig eth0 up

(root@kali)-[~]
#
```

User Management

Working as the root user all the time is very dangerous because one wrong command can break the system. So, I added a new user for daily tasks. I used this commands in terminal:

1. Add user: `sudo adduser internuser`

2. Assign admin: `sudo usermod -aG sudo internuser`

3. Switch user: `su - internuser`

4. Sudo whoami:- to verify admin access

```
newuser@kali: ~  
Session Actions Edit View Help  
(root@kali)-[/home/kali]  
# sudo adduser newuser  
New password:  
Retype new password:  
passwd: password updated successfully  
Changing the user information for newuser  
Enter the new value, or press ENTER for the default  
Full Name []: Jaydeep Joshi  
Room Number []: 1  
Work Phone []: 8160167347  
Home Phone []: --  
Other []: -  
Is the information correct? [Y/n] Y
```

```
(root@kali)-[/home/kali]  
# sudo usermod -aG sudo newuser  
  
(root@kali)-[/home/kali]  
# su - newuser  
(newuser@kali)-[~]  
$ sudo  
usage: sudo -h | -K | -k | -V  
usage: sudo -v [-ABkNnS] [-g group] [-h host] [-p prompt] [-u user]  
usage: sudo -l [-ABkNnS] [-g group] [-h host] [-p prompt] [-U user]  
[-u user] [command [arg ...]]  
usage: sudo [-ABbEHkNnPS] [-r role] [-t type] [-C num] [-D directory]  
[-g group] [-h host] [-p prompt] [-R directory] [-T timeout]  
[-u user] [VAR=value] [-i | -s] [command [arg ...]]  
usage: sudo -e [-ABkNnS] [-r role] [-t type] [-C num] [-D directory]  
[-g group] [-h host] [-p prompt] [-R directory] [-T timeout]  
[-u user] file ...  
  
(newuser@kali)-[~]  
$ sudo whoami  
[sudo] password for newuser:  
root
```

5. Documentation

Each task is essential in cybersecurity operations. Doing update keep system safe from new bugs. MAC spoofing hide our real physical device address on network which is important for privacy during testing. Good user management stop unauthorized peoples from getting root access.

6. Reflection

In conclusion, this project teach me how to setting up a professional cyber lab from scratch. I learned that virtualization is a safe, flexible way to practice hacking skills. This skills applies directly to real-world scenarios, like when we have to test a client's network. We must spoof our MAC so we don't trigger defensive alarms, and we must secure our own machines with proper user accounts before we start attacking others. It is very good preparation for my upcoming internship and my future career.